Jose Vargas
Fall 2026
Linux Administration
Prof: Adrianna Holden-Gouveia

Server Setup

Ubuntu Server Installation and Verification

Task

Install and configure Ubuntu Server on a virtual machine using QEMU/KVM on Fedora Linux 44. Configure remote access through SSH, verify network connectivity, and test file transfers using SCP.

Environment / Requirements

Host Environment

- Operating System: Fedora Linux 44 (KDE Plasma Desktop Edition)
- Virtualization: QEMU/KVM
- Virtual Machine Manager: virt-manager 5.1.0 with libvirt

Virtual Machine Configuration

- Operating System: Ubuntu Server 26.04.1 LTS
- CPU: 4 cores
- RAM: 8 GB
- Storage: 25 GB
- Network: NAT

Installation

Procedure

1. Download the Ubuntu Server ISO from the official Ubuntu website.

2. Open Virtual Machine Manager and create a new virtual machine using the downloaded ISO.

3. Allocate the CPU, RAM, and storage resources according to the host's capabilities. My host has 32 GB RAM, a 500 GB SSD, and a 12-core CPU.

4. During the Ubuntu Server installation, select the option to install OpenSSH Server to enable remote access.

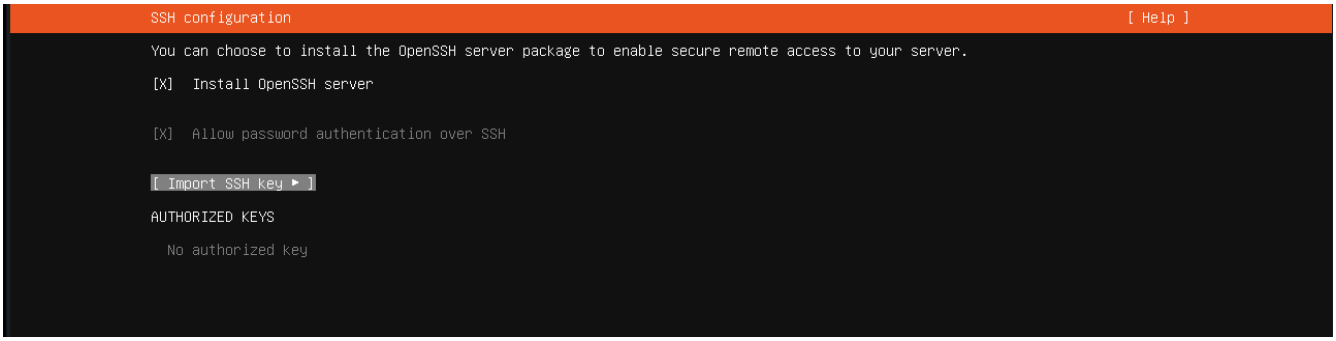
How it works

QEMU/KVM allows Ubuntu Server to run as a virtual machine on my Fedora host without requiring a separate physical computer. Virtual Machine Manager provides a graphical interface to create and configure the VM, including its CPU, memory, storage, and network settings.

The NAT network allows the virtual machine to access external networks through the host's virtual networking configuration.

OpenSSH Server is installed during the operating system setup to allow remote connections using the SSH protocol. This makes it possible to manage Ubuntu Server from another terminal without accessing the VM console directly.

Result



Network and Connectivity Verification

Network Interface and IP Address

Command

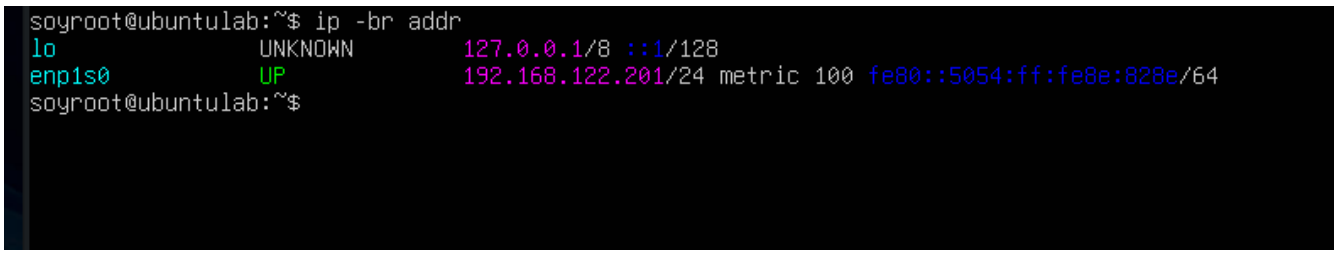
ip -br addr

How it works

The ip command is used to display and manage network information in Linux. The -br option displays the output in a brief format, while addr displays the IP addresses assigned to each network interface.

In the result, the enp1s0 interface is UP and has the IPv4 address 192.168.122.201/24, showing that the virtual machine has an active network interface with an assigned IP address.

Result



Internet Connectivity

Command

ping -c 3 google.com

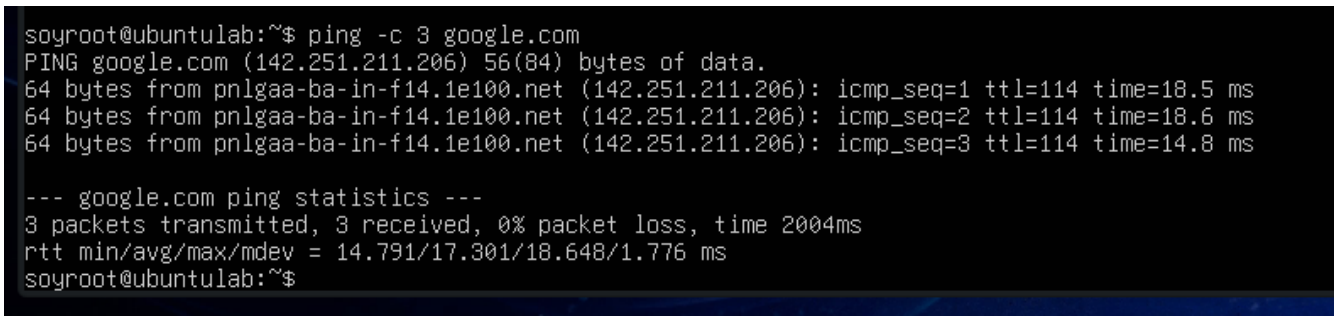
How it works

The ping command tests network connectivity by sending ICMP Echo Requests to a destination and waiting for replies. In this case, google.com is the destination.

The -c 3 option limits the test to three requests instead of allowing the command to continue until manually stopped.

The result shows three packets transmitted and three received, with 0% packet loss. This confirms that the server was able to resolve the hostname and communicate with the destination during the test.

Result



Remote Server Access

SSH Connection

Command

ssh soyroot@192.168.122.201

How it works

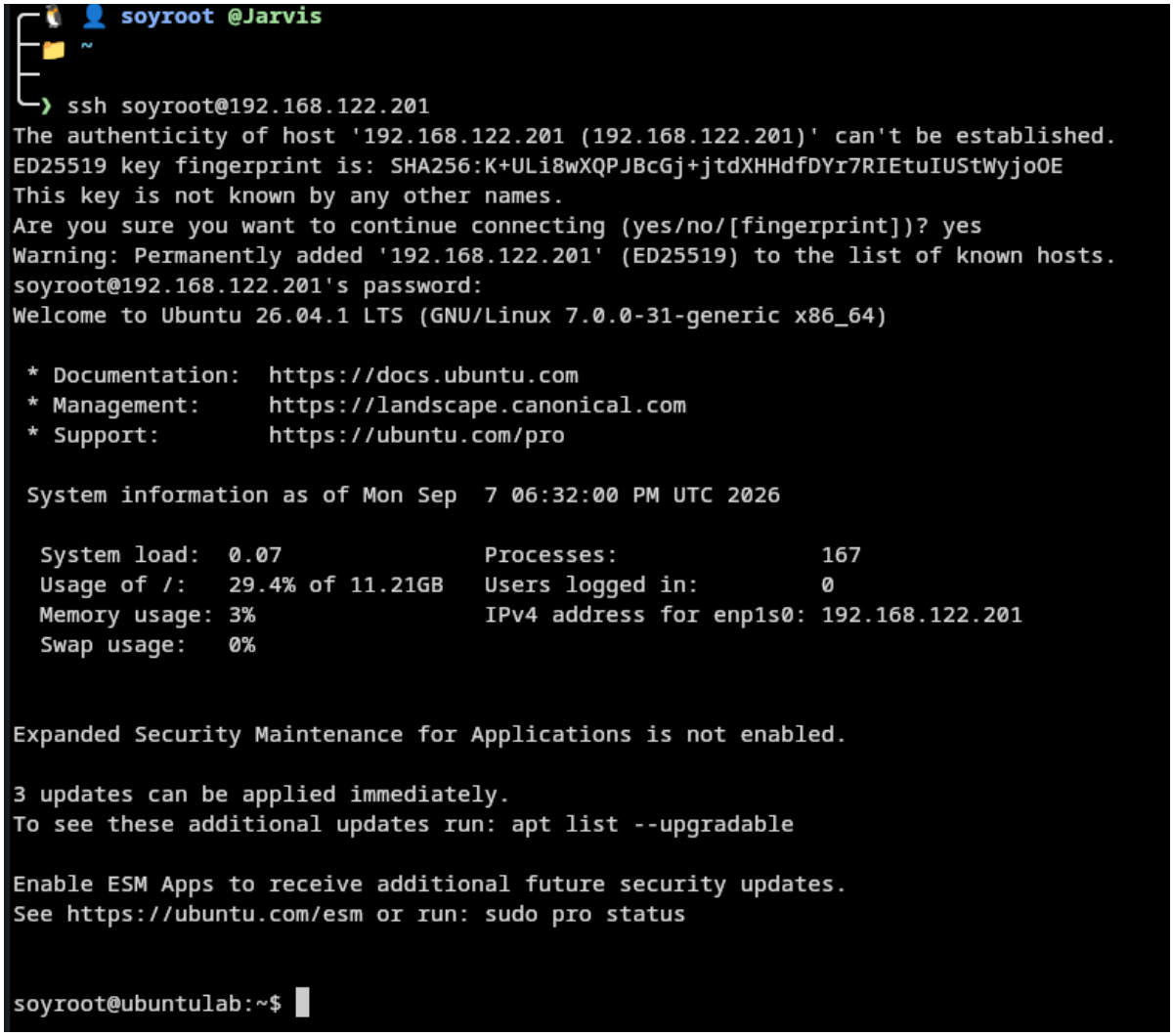
The ssh command establishes a secure remote terminal connection to another computer using the SSH protocol.

In this command, soyroot specifies the username used to log in to the remote server, while 192.168.122.201 is the server's IP address. The @ symbol separates the username from the destination address.

During the first connection, SSH asks to confirm the server's host key. After accepting the key and entering the user's password, the connection is established.

The result shows that I successfully connected to Ubuntu Server from my Jarvis terminal and obtained a remote shell session.

Result



File Transfer Verification

SCP File Transfer

Command

scp scp-test.txt soyroot@192.168.122.201:~/

How it works

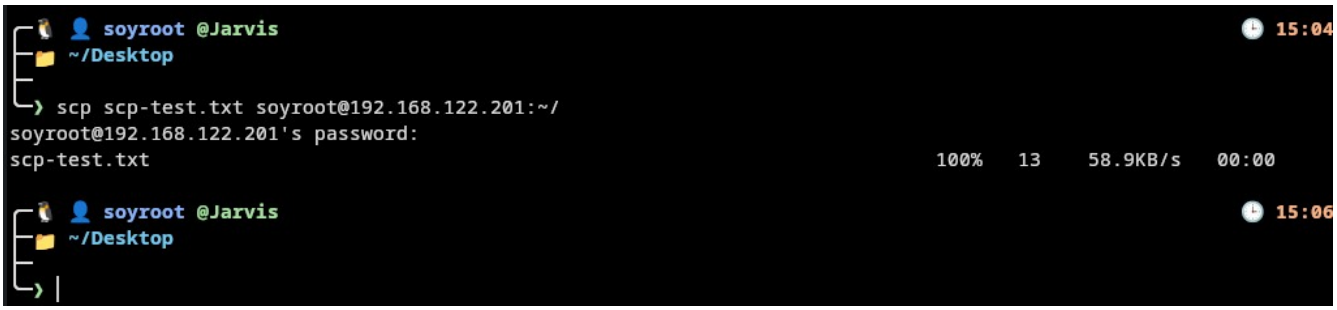
The `scp` command is used to securely transfer files between computers over an SSH connection.

In this command, scp-test.txt is the local file being transferred. soyroot@192.168.122.201 identifies the remote username and server IP address.

The colon (:) separates the remote destination from its file path, while ~/ specifies the remote user's home directory.

After authentication, SCP transfers the file to the destination. The terminal output shows that the transfer reached 100%.

Result



Ubuntu Server Results

The Ubuntu Server virtual machine was installed and configured with OpenSSH Server. Network verification showed an active interface with an assigned IP address, and the connectivity test completed with 0% packet loss.

Remote SSH access was established successfully, and the SCP transfer completed according to the terminal output.

CentOS Stream 10 Installation and Verification

Task

Install and configure CentOS Stream 10 on a virtual machine using QEMU/KVM on Fedora Linux 44. Configure the server and verify its network connectivity, remote access, and file transfer capabilities.

Environment / Requirements

Virtual Machine Configuration

- Operating System: CentOS Stream 10
- CPU: 4 cores
- RAM: 8 GB
- Storage: 25 GB
- Network: NAT

The CentOS virtual machine uses the same host environment and virtualization software described in the Ubuntu Server installation.

Installation

Procedure

1. Download the CentOS Stream 10 ISO from the official CentOS website.
2. Open Virtual Machine Manager and create a new virtual machine using the CentOS Stream 10 ISO.
3. Allocate the CPU, RAM, and storage resources according to the virtual machine requirements.
4. During installation, select **Installation Destination** to configure the storage device.
5. Select the appropriate virtual disk for the operating system installation.
6. Configure the user account and administrator account.
7. Complete the installation process and restart the virtual machine.

How it works

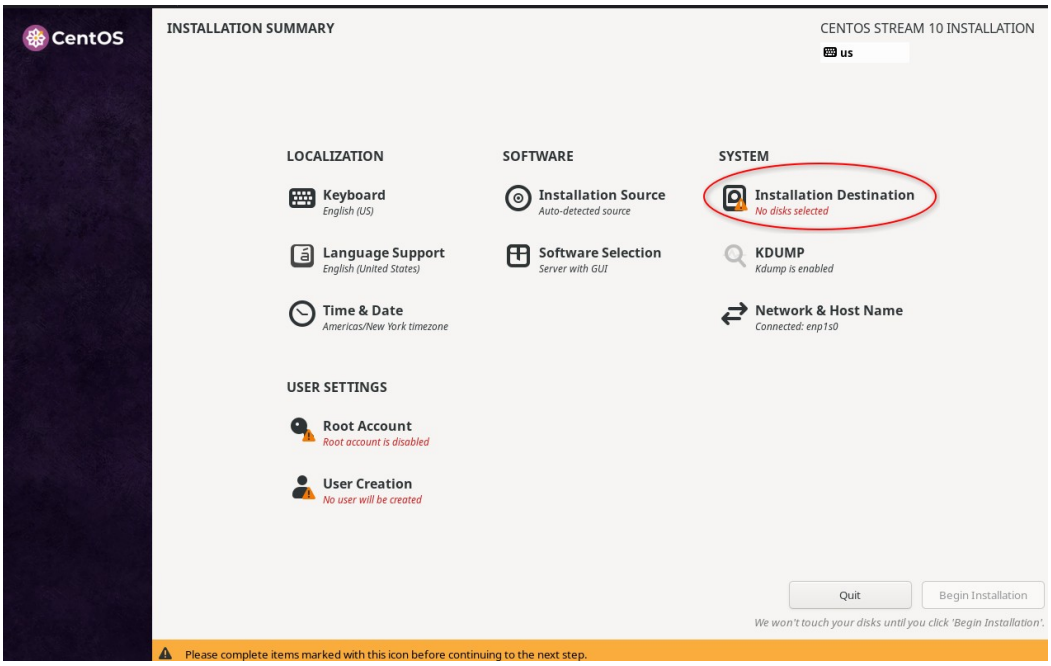
The CentOS Stream virtual machine follows the same virtualization process used for Ubuntu Server. QEMU/KVM provides the virtualized hardware, while Virtual Machine Manager is used to configure the CPU, memory, storage, and network resources.

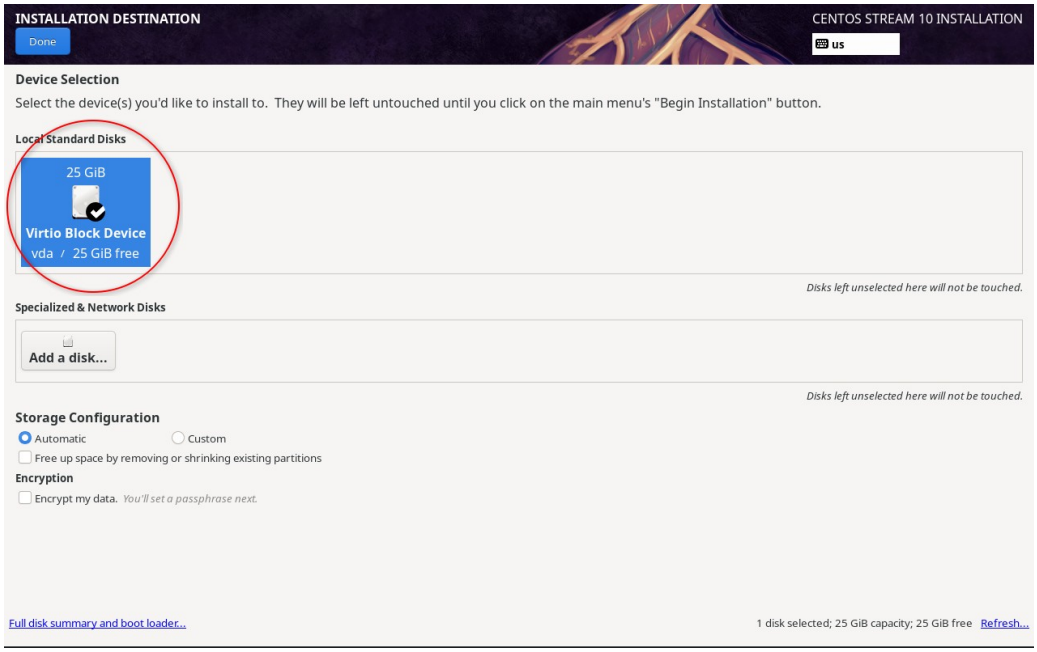
CentOS Stream uses a different installation interface from Ubuntu Server. During installation, the Installation Destination section allows us to select the virtual disk where the operating system will be installed.

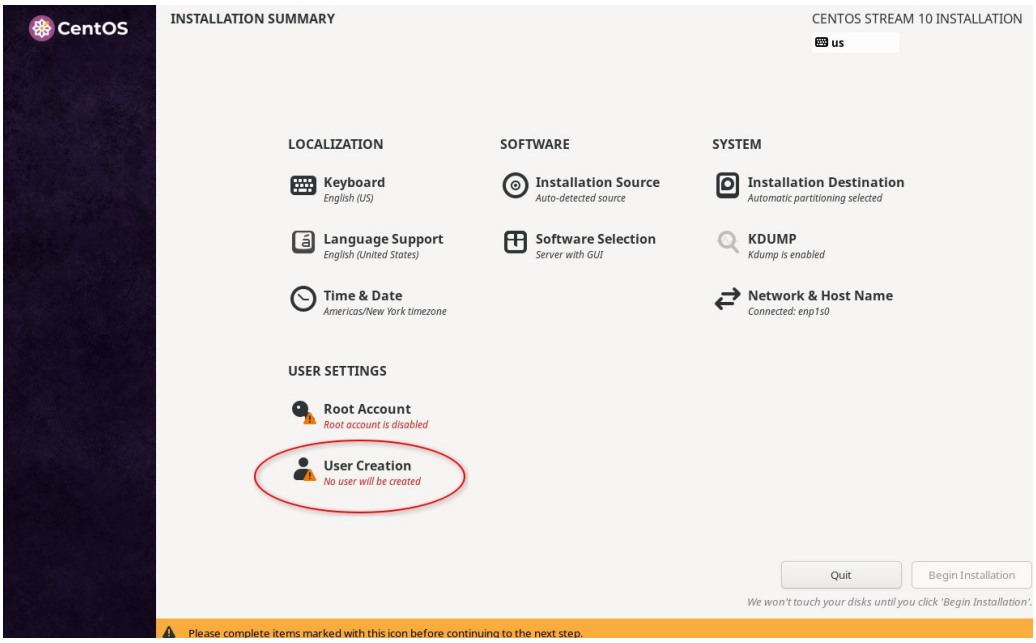
The selected 25 GB virtual disk is used as the installation destination. The installer also provides options to configure user accounts and administrative access.

Once the installation is complete, the virtual machine can boot into CentOS Stream independently of the Ubuntu Server virtual machine.

Result


CentOS
INSTALLATION SUMMARY
CENTOS STREAM 10 INSTALLATION
us
LOCALIZATION
Keyboard
English (US)
Language Support
English (United States)
Time & Date
Americas/New York timezone
SOFTWARE
Installation Source
Auto-detected source
Software Selection
Server with GUI
SYSTEM
Installation Destination
No disks selected
KDUMP
Kdump is enabled
Network & Host Name
Connected: enp1s0
USER SETTINGS
Root Account
Root account is disabled
User Creation
No user will be created
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.
Please complete items marked with this icon before continuing to the next step.


INSTALLATION DESTINATION
Done
CENTOS STREAM 10 INSTALLATION
us
Device Selection
Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.
Local Standard Disks
25 GiB
Virtio Block Device
vda / 25 GiB free
Disks left unselected here will not be touched.
Specialized & Network Disks
Add a disk...
Disks left unselected here will not be touched.
Storage Configuration
Automatic
Custom
Free up space by removing or shrinking existing partitions
Encryption
Encrypt my data. You'll set a passphrase next.
Full disk summary and boot loader...
1 disk selected; 25 GiB capacity; 25 GiB free
Refresh...



Network and Connectivity Verification

The same verification procedures used for Ubuntu Server can also be applied to CentOS Stream.

Network Interface and IP Address

Command

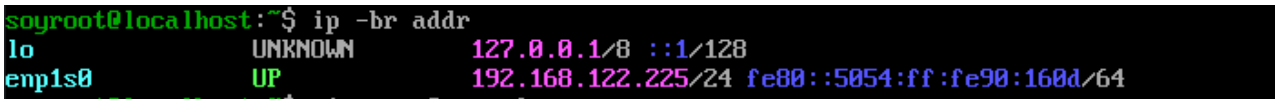
ip -br addr

How it works

The ip command displays information about network interfaces. The -br option produces a brief output, while addr displays the IP addresses assigned to the interfaces.

This command can be used to identify the active network interface and verify that CentOS Stream has an assigned IP address.

Result



Internet Connectivity

Command

ping -c 3 google.com

How it works

The ping command sends ICMP Echo Requests to google.com to test network connectivity. The -c 3 option limits the test to three requests.

The output shows whether the destination responds, the number of packets transmitted and received, and any packet loss.

Result



Remote Server Access

SSH Connection

Command

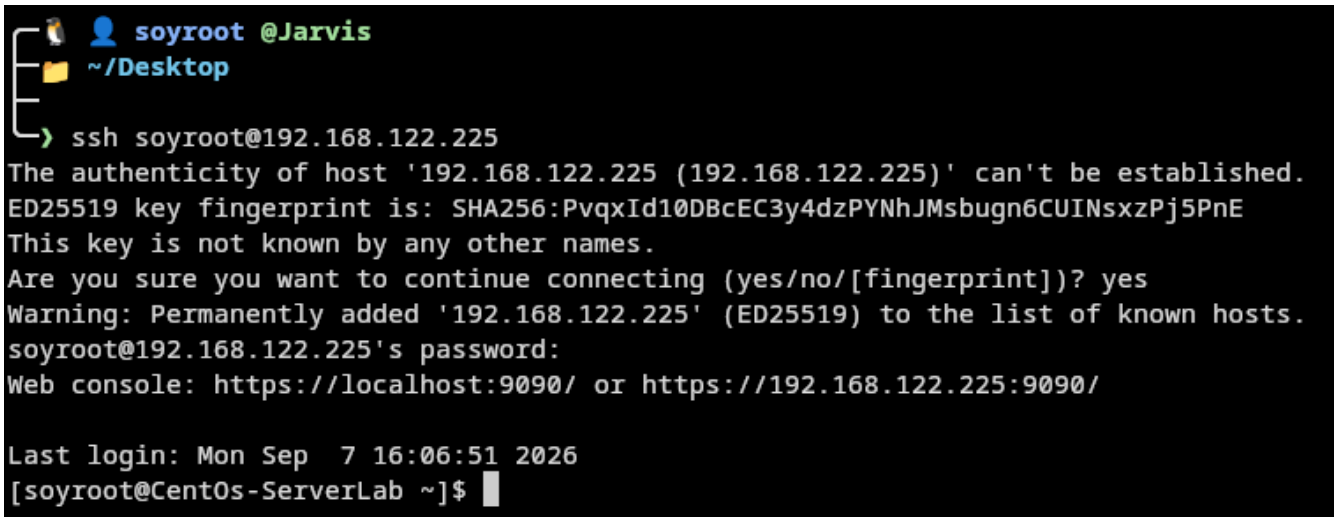
ssh soyroot@192.168.122.225

How it works

The `ssh` command establishes a secure remote terminal connection to CentOS Stream. The username identifies the account used for authentication, while the IP address identifies the destination server.

Once the connection is authenticated, the remote shell allows us to manage CentOS Stream without using the virtual machine's graphical console.

Result



File Transfer Verification

SCP File Transfer

Command

scp scp-test.txt soyroot@192.168.122.225:~/

How it works

The scp command transfers a file securely using SSH.

In this example, scp-test.txt is the local file being transferred. The username and IP address identify the remote CentOS server, while :~/ specifies the destination user's home directory.

After the transfer, the file can be verified on CentOS Stream by checking that it exists in the destination directory.

Result





CentOS Stream Results

The CentOS Stream 10 virtual machine was successfully installed and configured using QEMU/KVM and Virtual Machine Manager.

The network interface was UP with the IPv4 address 192.168.122.225/24. The internet connectivity test completed with three packets transmitted and received, resulting in 0% packet loss.

Remote access through SSH was successfully established, and the SCP file transfer completed with 100% progress. The transferred file was also verified on the CentOS server using the ls command.

Sources

• Ubuntu. Download Ubuntu Server.

• https://ubuntu.com/download/server

• CentOS. Download.

• https://www.centos.org/download/